

1. PROOF

Let $\mathcal{D}$ be the top-aligned 5-by-5 Ferrers diagram with column heights $(c_1, \dots, c_5) = (3, 3, 5, 5, 5)$. Write $\kappa_F(\mathcal{D}, 5)$ for the largest dimension of an F -linear space of matrices supported on $\mathcal{D}$ in which every nonzero matrix has rank 5. The Etzion–Silberstein bound is

$$\nu_{\min}(\mathcal{D}, 5) = \min_{0 \leq j < 5} \sum_{i=1}^{5-j} \max\{0, c_i - 4 + j\}. \quad (1.1)$$

Theorem 1.1 (Etzion–Silberstein counterexample). *For $F = \mathbb{F}_{169}$,*

$$\kappa_F(\mathcal{D}, 5) = 2 < 3 = \nu_{\min}(\mathcal{D}, 5).$$

Consequently, the Etzion–Silberstein bound is not universally attained.

Proof. The five quantities minimized in (1.1) are

$$(3, 4, 5, 4, 3),$$

so the bound is 3.

We first prove that the optimum is at most 2. Put $q = 169$ and suppose that a three-dimensional code exists. Choose a basis M_0, M_1, M_2 , put $M(x) = \sum_{i=0}^2 x_i M_i$, and identify x with a column vector. The Ferrers support gives

$$M(x) = \begin{pmatrix} P(x) & B(x) \\ 0 & D(x) \end{pmatrix}, \quad P(x) = [A_1 x, A_2 x],$$

where $A_1, A_2 \in M_3(F)$. Each A_i is invertible, since otherwise some nonzero $M(x)$ has a zero column. Moreover, $T = A_1^{-1} A_2$ has no eigenvalue in F , since an eigenvector would make the first two columns of $M(x)$ dependent. Thus the cubic characteristic polynomial of T is irreducible. Over $E = \mathbb{F}_{q^3}$ it has an eigenvector $v \neq 0$, and hence

$$\det M(v) = 0. \quad (1.2)$$

Now set $f(X) = \det(\sum_{i=0}^2 X_i M_i)$. This is a homogeneous quintic with no nonzero zero over F . It is irreducible over F : otherwise one of its homogeneous factors has degree at most 2, and such a form in three variables has a nonzero zero over a finite field, by elementary linear algebra in degree 1 and Chevalley–Warning in degree 2.

If f were absolutely irreducible, the normalization of the plane curve $f = 0$ would have genus at most 6. Hasse–Weil would give at least $169 + 1 - 12 \cdot 13 = 14$ rational points; their images would be projective F -zeros of f , a contradiction. Hence f is not absolutely irreducible.

Since $\text{char } F = 13 > 5$, the quintic is geometrically reduced. Frobenius acts transitively on its absolutely irreducible factors, whose number times their common degree is 5. Irreducibility over F and failure of absolute

irreducibility therefore force five conjugate lines. Thus, for $L = \mathbb{F}_{q^5}$ and a nonzero $c \in F$,

$$f = c \prod_{i=0}^4 \ell^{(q^i)}, \quad \ell = a_0 X_0 + a_1 X_1 + a_2 X_2, \quad a_i \in L. \quad (1.3)$$

The coefficients a_0, a_1, a_2 are F -linearly independent. Otherwise ℓ , and therefore f , would have a nonzero F -zero. Because $E \cap L = F$ (the extension degrees 3 and 5 are coprime), they remain independent over E . The same holds for their Frobenius conjugates, so (1.3) has no nonzero zero in E^3 . This contradicts (1.2). No three-dimensional code exists, and a code of larger dimension would contain one; hence $\kappa_F(\mathcal{D}, 5) \leq 2$.

For the reverse inequality, let $K = \mathbb{F}_{q^5} = F(\alpha)$ and use the power basis $1, \alpha, \dots, \alpha^4$. For $a, b \in F$, let $M_{a+b\alpha}$ be the matrix of multiplication by $a + b\alpha$ on K . The space

$$C_0 = \{M_{a+b\alpha} : a, b \in F\}$$

has dimension 2. Its first two columns represent $a + b\alpha$ and $a\alpha + b\alpha^2$, so their last two entries vanish; there is no restriction on the remaining columns. Thus C_0 is supported on $\mathcal{D}$. Every nonzero member is invertible because it is multiplication by a nonzero field element. Therefore $\kappa_F(\mathcal{D}, 5) \geq 2$, completing the proof. $\square$